

Password Generator (EASY Version)

Understand-

Inputs: how many letters
 how many symbols
 how many numbers

Outputs: the user's password

Constraints: symbols are limited to list of allowed symbols

Rules: password is # of letters + # of symbols + # of numbers long

START

```
SET LETTERS = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z', 'A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z']
```

```
SET NUMBERS = ['1', '2', '3', '4', '5', '6', '7', '8', '9', '0']
```

```
SET SYMBOLS = ['!', '#', '$', '%', '&', '(', ')', '*', '+']
```

```
SET password = ""
```

```
SET character = ""
```

```
SET min_range = 2
```

```
SET max_range = 8
```

```
FUNCTION random_choice_from_list(input_list)
```

```
    RETURN random_choice
```

```
FUNCTION is_integer(input_string)
```

```
    IF input_string is integer THEN
```

```
        RETURN True
```

```
    ELSE
```

```
        OUTPUT "Please enter an integer!"
```

```
        RETURN False
```

```
    END IF
```

```
FUNCTION validate_number_range(input_number, min_range, max_range)
```

```
    If min_range <= input_number <= max_range THEN
```

```
        RETURN True
```

```
    ELSE
```

```
        OUTPUT "Enter a number between (min_range) and (max_range)"
```

```
        RETURN False
```

```
WHILE True
```

```
    INPUT "How many 'letters' should your password have?"
```

```
    SET num_letters
```

```
    IF is_integer(num_letters) THEN
```

```
        num_letters = int(num_letters)
```

```
        IF validate_number_range(num_letters, min_range, max_range)
```

```
            BREAK
```

```
WHILE True
```

```
    INPUT "How many 'symbols' should your password have?"
```

```
    SET num_symbols
```

```
    IF is_integer(num_symbols) THEN
```

```
        num_symbols = int(num_symbols)
```

```
        IF validate_number_range(num_symbols, min_range, max_range)
```

```
            BREAK
```

WHILE True

INPUT "How many 'numbers' should your password have?"

SET num_numbers

IF is_integer(num_numbers) THEN

num_numbers = int(num_numbers)

IF validate_number_range(num_numbers, min_range, max_range)

BREAK

SET i = 1

IF i <= num_letters THEN

character = random_choice_from_list(LETTERS)

password = password append character

i = i + 1

END IF

character = ""

SET i = 1

IF i <= num_symbols THEN

character = random_choice_from_list(SYMBOLS)

password = password append character

i = i + 1

END IF

character = ""

SET i = 1

IF i <= num_numbers THEN

character = random_choice_from_list(NUMBERS)

password = password append character

i = i + 1

END IF

OUTPUT "Your new password is : (password)"

